Field Switching in Learning Parity with Noise

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Learning Parity with Noise (LPN)

computationally indistinguishable

$$(A, As + e) \approx_c (A, u)$$

$A \sim \text{Uniform}(\mathbb{F}_2^{m \times n})$

$s \sim \text{Uniform}(\mathbb{F}_2^{n})$

$e_i \sim \text{Bernoulli}(\tau)$ with parameter μ, i.i.d.

$u \sim \text{Uniform}(\mathbb{F}_2^{m})$

LPN over Finite Fields

$$(A, As + e) \approx_c (A, u)$$

$A \sim \text{Uniform}(F_q^{m \times n})$

$s \sim \text{Uniform}(F_q^n)$

$e_i \sim \text{Bernoulli}(\tau)$ with parameter μ, i.i.d.

$u \sim \text{Uniform}(F_q^m)$

Bernoulli Noise over F_q

For $0 \le \mu \le 1$ and any finite field F_q

$$e \leftarrow \begin{cases} 0 & \text{with probability } 1 - \mu \\ \text{Uniform}(F_q) & \text{with probability } \mu \end{cases}$$

Why study hardness of LPN over F_q?

Secure Multiparty Computation (MPC) Applications

Pseudorandom Correlation Generators (PCGs) requires hardness assumptions over its underlying field.

Theory

Understand how field affects the hardness of LPN.

LPN Variants in PCG Applications

Field choice

Applications often need LPN over that field: $\mathbb{F}_p$−LPN , $\mathbb{F}_{p^n}$-LPN

Communication complexity: noise sharing is a bottleneck

Two parties must privately share nonzero LPN error locations.
Ad hoc assumptions, e.g. **stationary LPN**, cut this cost.

Efficiency

Ring LPN and its multivariate generalization **QA-SD** can give further speedups.

LPN Variants in PCG Applications

$\mathbb{F}_p$−LPN , $\mathbb{F}_{p^n}$-LPN

Stationary LPN	Sample k instances A_j, $A_j s_j + e_j$ for $j \in [k]$ where across instances e_j is nonzero in the same locations.
Galois-Ring LPN	Elements are in $F_{p^m}[x]/f(x)$ where f(x) is an irreducible polynomial.
Quasi-abelian syndrome (QA-SD) decoding (i.e. Module LPN)	Elements are in $F[x_1, \ldots, x_r]/(x_1^d-1, \ldots, x_r^d-1)$

Prior Work

Problem	Prior work
$\mathbb{F}_p$-LPN	Standard assumption; conjectured subexponential security for suitable noise rates.
$\mathbb{F}_{p^n}$-LPN	[Unal23] Reduction to and from $\mathbb{F}_p$-LPN
Stationary LPN	[KPRR25] Security against linear attacks.
Galois-ring LPN	[LXYY25b] $\mathbb{F}_{p^n}$-to-$\mathbf{GR}(p^m, n)$ reduction with exponential noise loss.
QA-SD	[BCCD23] Search-to-decision for specific instantiations in the low-rate regime.

This Work

	Hardness at noise μ follows from:
Stationary LPN	$\mathbb{F}_p$-LPN, noise $O(\mu/k)$ k stationary instances over $\mathbb{F}_p$.
Galois-Ring LPN	Z_{p^m}-invertible noise LPN, noise $\approx \mu/n$ Standard noise if p^n is superpoly.
QA-SD over $\mathbb{F}_p$	QA-SD over $\mathbb{F}_{p^n}$, noise μ

General Technique

Start with standard re-randomization technique used in Learning with Errors (LWE)/LPN reductions

Start with original instance

$$A \quad , \quad As + e$$

Transform to new instance: multiply by a random matrix R

$$RA \quad , \quad RAs + Re$$

random

Hope: noise is distributed according to desired distribution for new instance

Trace: linear function from $\mathbb{F}_{p^n}$ to $\mathbb{F}_p$

$$\mathrm{Tr}(x) = x + x^p + x^{p^2} + \cdots + x^{p^{n-1}}$$

Crucial properties:

1) $\mathbb{F}_p$-Linearity: $\mathrm{Tr}(ax) = a \cdot \mathrm{Tr}(x)$ if $a \in \mathbb{F}_p$

2) $\mathrm{Tr}(0) = 0$

3) Tr(uniform distribution) = uniform distribution

Reduction: $\mathbb{F}_{p^n}$ to $\mathbb{F}_p$

$$\mathrm{Tr}(A) \quad , \quad \mathrm{Tr}(As + e) \quad = \quad \mathrm{Tr}(As) \quad + \quad \mathrm{Tr}(e)$$

$\mathrm{Tr}(As)$: can be simplified

$\mathrm{Tr}(e)$: this is uniform w.p. μ and 0 w.p. $1 - \mu$

Prior Work [Unal23]

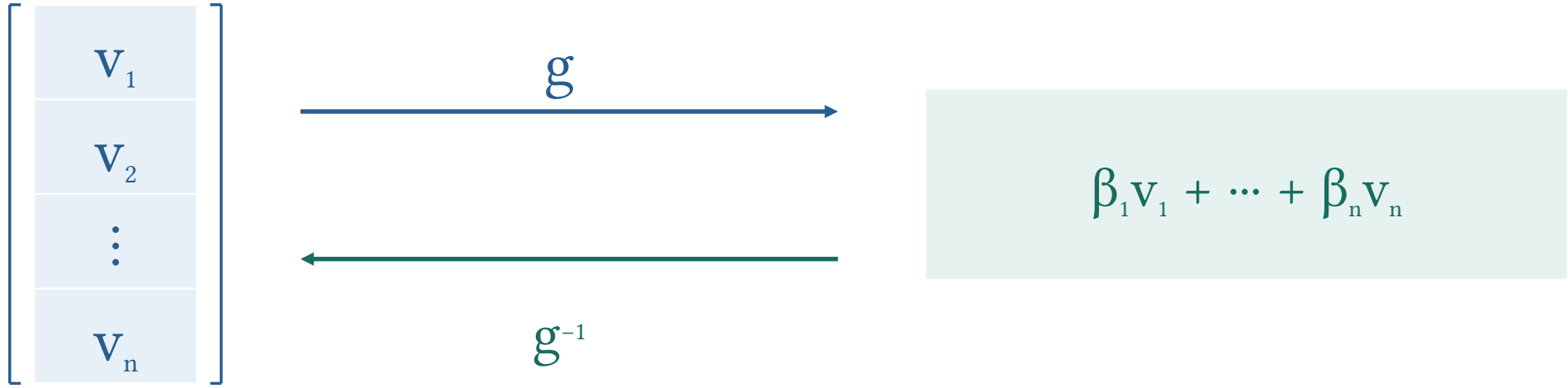


Gadget g, g^{-1}

Fix a basis $g = (\beta_1, \beta_2, \ldots, \beta_n)$ of F_{p^n} over F.

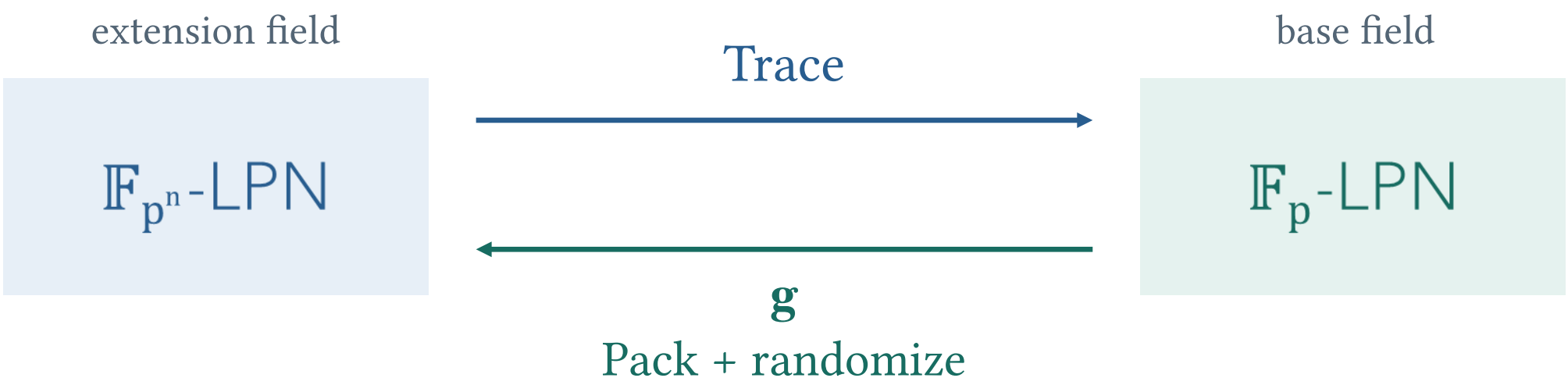
n F-coordinates

One F_{p^n}-element



Both g and g^{-1} are linear maps

Prior Work [Unal23]



Why not use g^{-1} instead of Trace?

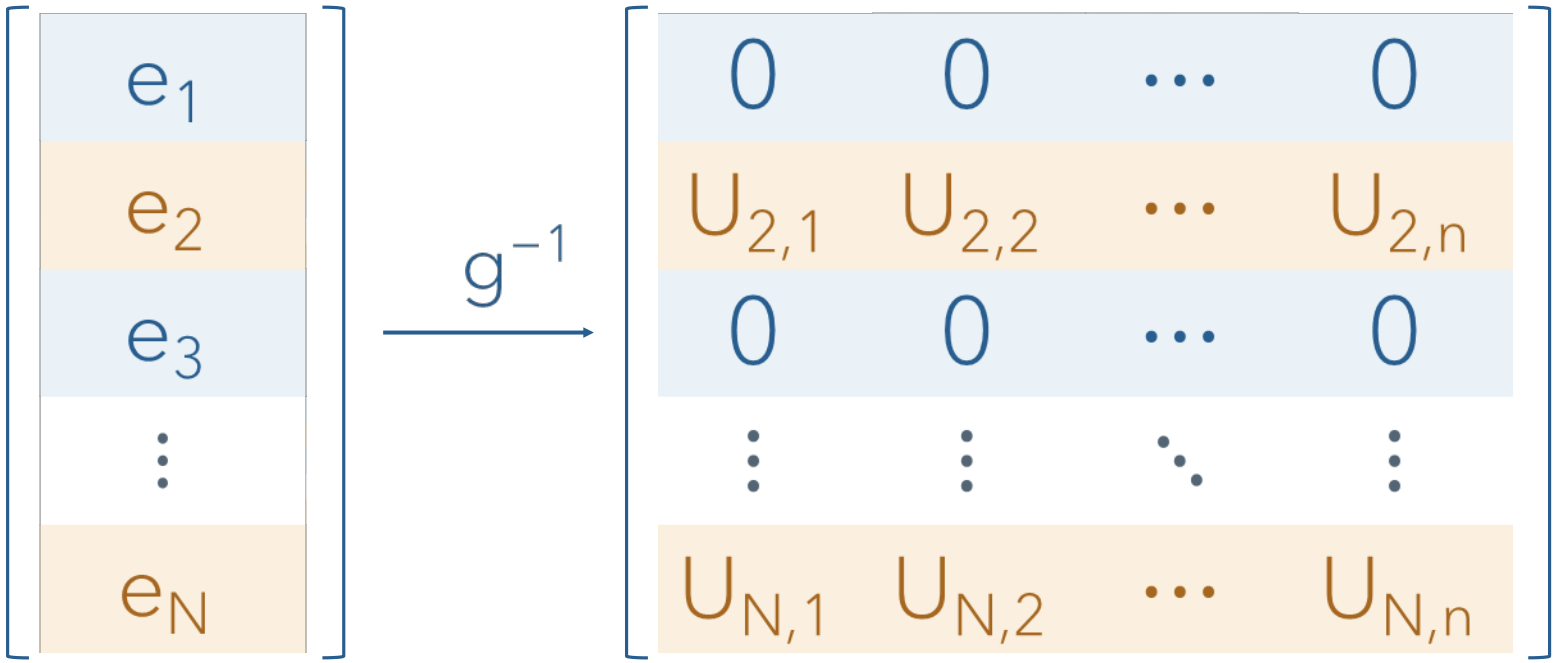
Reduction: $\mathbb{F}_{p^n}$ to $\mathbb{F}_p$ with g^{-1}

$$g^{-1}(A)\ ,\quad g^{-1}(As + e) \;=\; g^{-1}(As) \;+\; g^{-1}(e)$$

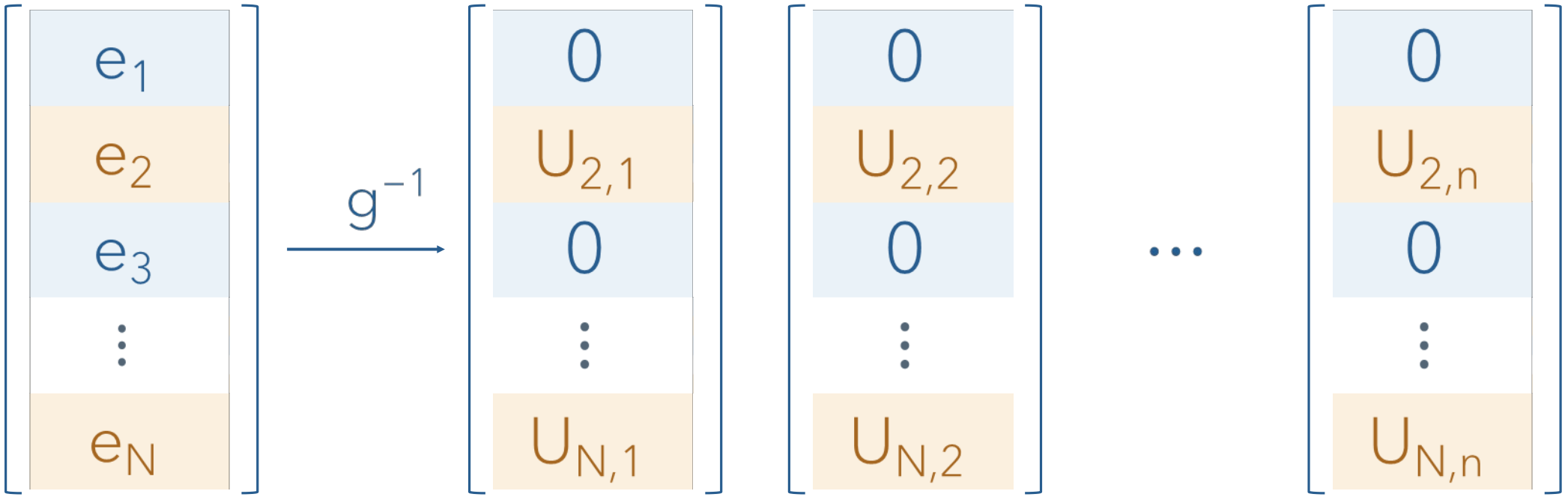
can be simplified

Is this noise distribution Bernoulli?

Noise Distribution: $g^{-1}(e)$

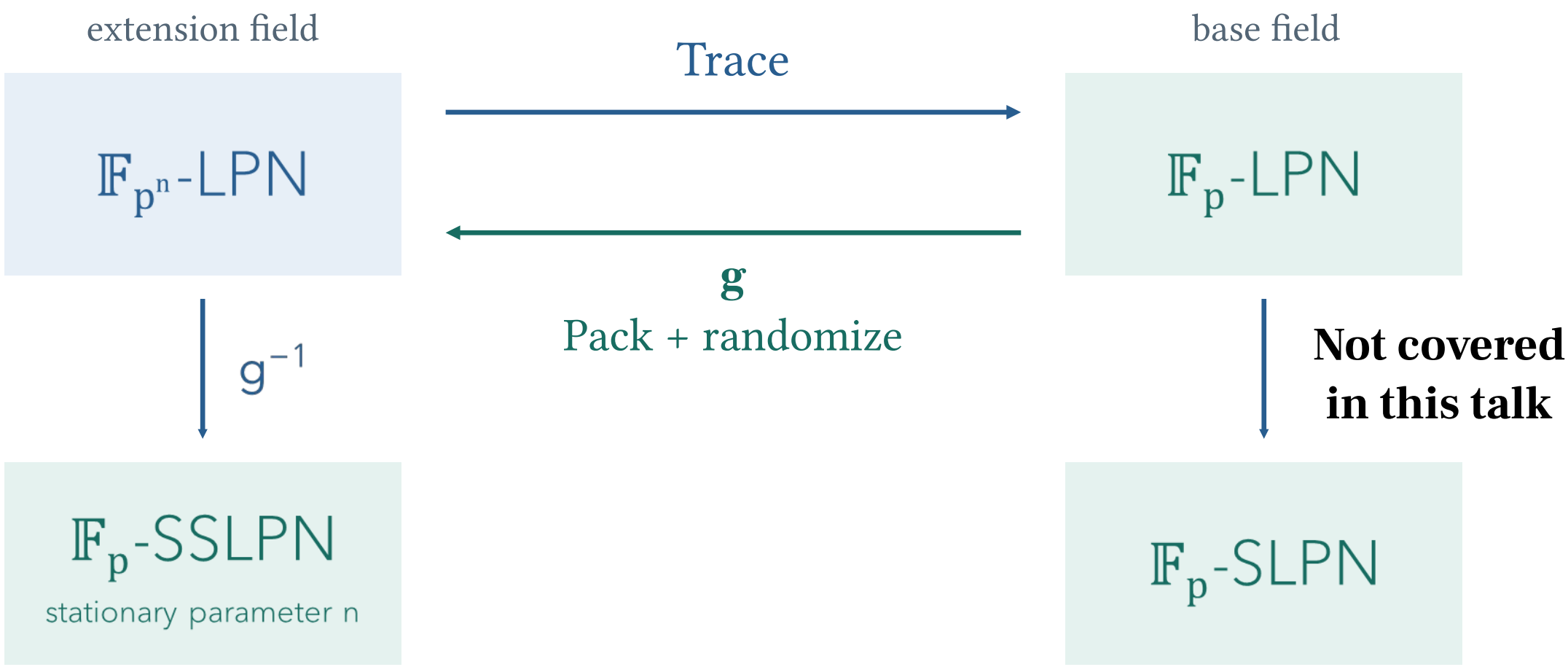


Noise Distribution: $g^{-1}(e)$

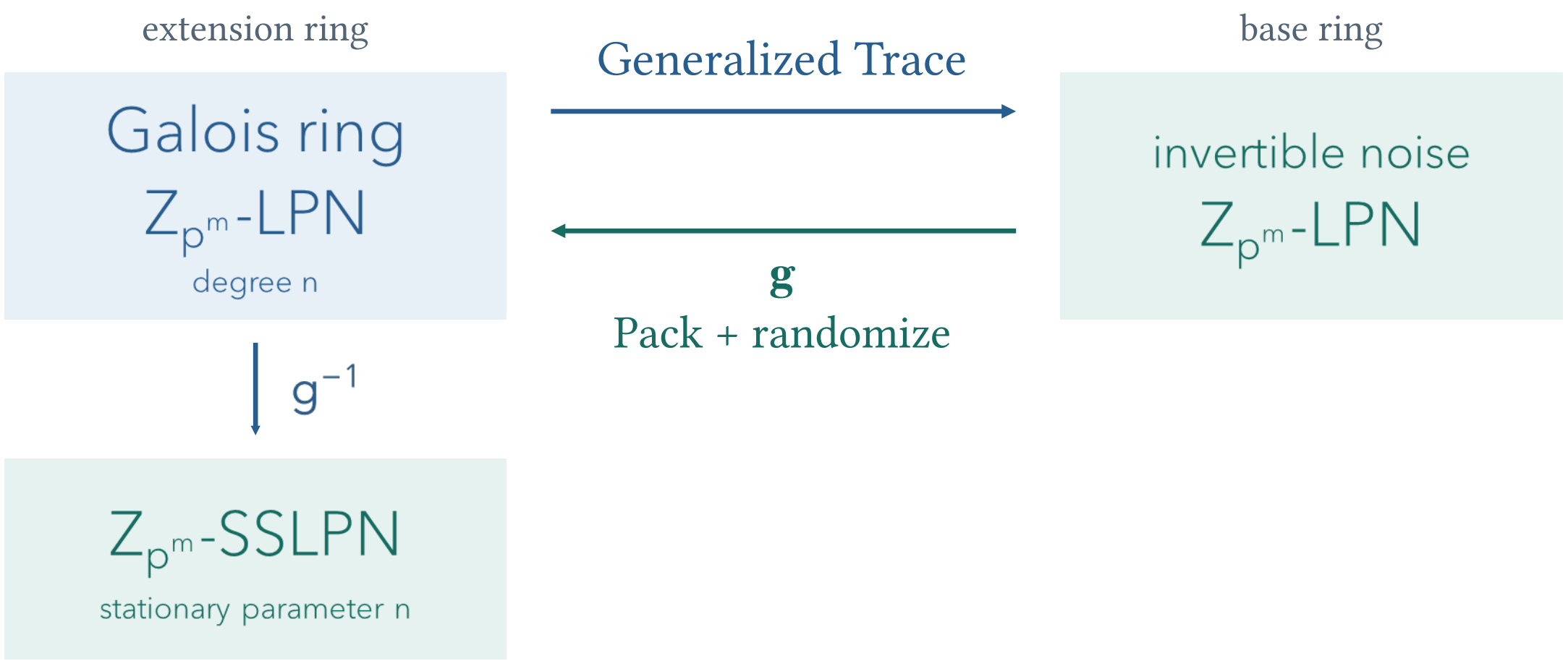


This gives n error vectors matching the stationary distribution!

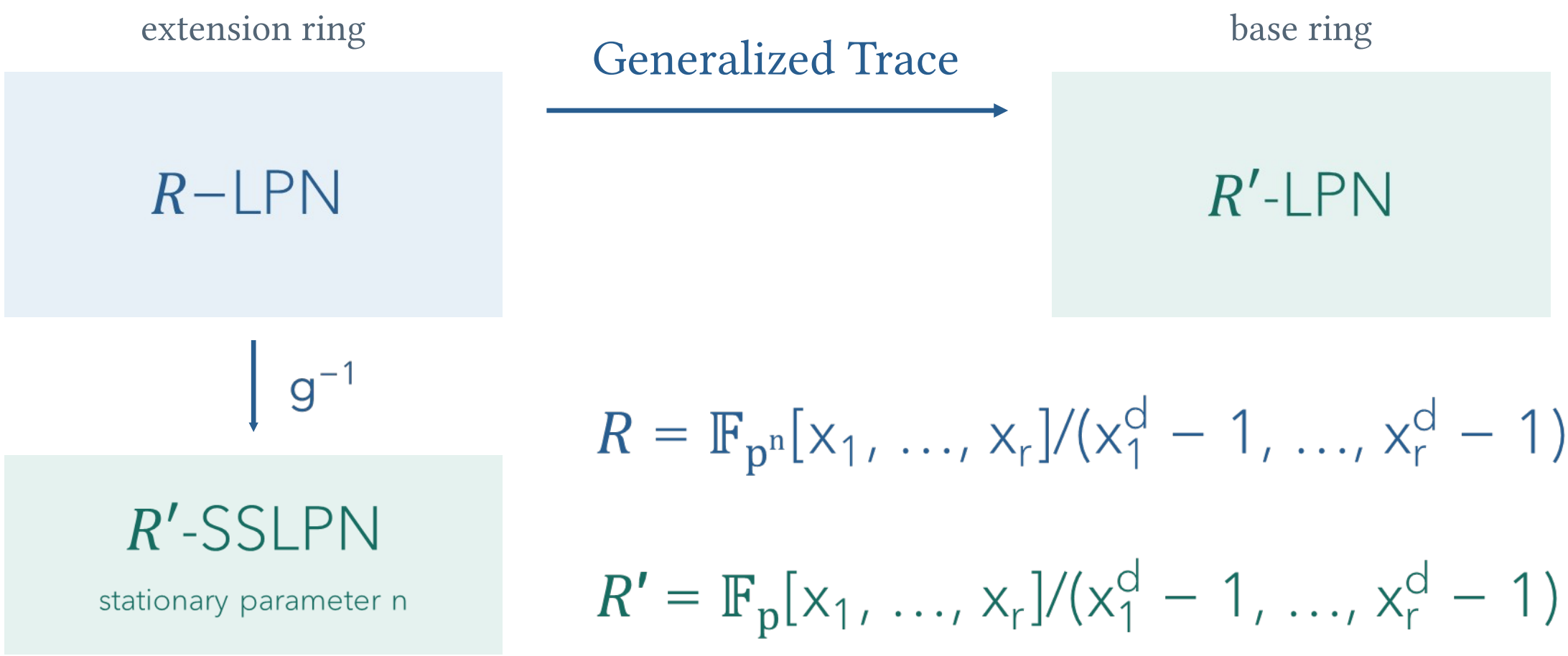
Summary: LPN Reductions



Additional Results: Galois rings



Additional Results: QA-SD



$$R = \mathbb{F}_{p^n}[x_1, \ldots, x_r]/(x_1^d - 1, \ldots, x_r^d - 1)$$

$$R' = \mathbb{F}_p[x_1, \ldots, x_r]/(x_1^d - 1, \ldots, x_r^d - 1)$$

Open Questions

1. Reduce QA-SD over $\mathbb{F}_{p^n}$ to QA-SD over $\mathbb{F}_p$

2. Reductions from $\mathbb{F}_p$-LPN to Galois-ring LPN and QA-SD LPN

Thank you!